

The Long-Range $O(N)$ Question in arXiv:2009.07209

Directly answered by arXiv:2310.01697

Literature-status identification and proof audit

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Status and exact source question

The status is **literature_already_answered**. The resolving work is a 2023 preprint, and the exact implication needed by the source can also be checked directly from the corrected version of Kloeckner’s coupling theorem.

In the final paragraph of Example 4.2 of Cioletti–Melo–Ruviano–Silva, *On the dimension of the space of harmonic functions on transitive shift spaces*, arXiv:2009.07209 (published in *Advances in Mathematics* 385 (2021), 107758), the authors consider

$$X = (\mathbb{S}^{N-1})^{\mathbb{N}}, \quad f(x) = \sum_{n=1}^{\infty} J(n) \langle x_1, x_{n+1} \rangle, \quad |J(n)| = O(n^{-2-\varepsilon}), \quad \varepsilon > 2. \quad (2)$$

They explain that their functional central limit theorem (FCLT) argument would extend from scalar spins to this uncountable alphabet if one could prove, for centered observables in the relevant logarithmic Hölder class,

$$\|\mathcal{L}_{\bar{f}}^n \varphi\|_{\infty} \leq D n^{1-\varepsilon}. \quad (3)$$

They identify the missing ingredient as an uncountable-alphabet extension of the optimal-transport method of Kloeckner.

Direct later answer

Silva–Mesquita–Matias, *Some Probabilistic Properties of General Topological Markov Chains*, arXiv:2310.01697 (2023), explicitly cites the 2021 paper and revisits this task. Its Section 5 first formulates the natural coupling for general alphabets, then proves polynomial decay for compact metric alphabets, and finally applies the result to a long-range $O(N)$ model. For the canonical interaction

$$f(x) = \sum_{n=1}^{\infty} \frac{\langle x_1, x_{n+1} \rangle}{n^{2+\varepsilon}},$$

the paper obtains (3), constructs a solution of the Poisson equation, and concludes the FCLT. Thus this is a separate later-paper answer, not a same-paper resolution.

The later paper writes the canonical coefficients in its model subsection, while the original question permits every sequence satisfying the bound in (2). The proof below verifies that the same argument

covers that full coefficient class. It also makes explicit two formula-level corrections: a coordinate projection is not uniformly Lipschitz for the product metric, and the decay estimate is proportional to an observable norm, not to a potential norm.

Audited form of the answer

Endow X with

$$d(x, y) = \sum_{k=1}^{\infty} 2^{-k} \min\{\|x_k - y_k\|, 1\}. \quad (4)$$

For $\gamma > 0$, write

$$\omega_\gamma(r) = (\log(r_0/r))^{-\gamma}$$

near zero, extended in the standard way to a modulus of continuity on the diameter of X . Denote the corresponding logarithmic Hölder space by $C^\gamma \log(X)$.

Let p be the a priori probability on $\mathbb{S}^{N-1}$ and let

$$\mathcal{L}_f u(x) = \int_{\mathbb{S}^{N-1}} e^{f(ax)} u(ax) dp(a).$$

The equilibrium measure and normalized potential below are those furnished by the Ruelle–Perron–Frobenius normalization.

Audited answer. Assume (2). There are a positive eigenfunction h , an eigenvalue $\rho > 0$, and the normalized potential

$$\bar{f} = f + \log h - \log h \circ \sigma - \log \rho$$

such that, for every $\varphi \in C^{(\varepsilon-1)\log(X)}$ satisfying $\int_X \varphi d\mu_f = 0$,

$$\|\mathcal{L}_{\bar{f}}^n \varphi\|_\infty \leq C n^{-(\varepsilon-1)} \|\varphi\|_{(\varepsilon-1)\log}, \quad n \geq 1. \quad (5)$$

In particular the Poisson series converges and the FCLT criterion used in *arXiv:2009.07209* applies.

Proof. Put $r = d(x, y)$. From (4), and from the fact that two sphere points have distance at most 2,

$$\|x_k - y_k\| \leq 2 \min\{1, 2^k r\}. \quad (6)$$

The inner-product difference is bounded by

$$|\langle x_1, x_{n+1} \rangle - \langle y_1, y_{n+1} \rangle| \leq \|x_1 - y_1\| + \|x_{n+1} - y_{n+1}\|.$$

Using (6), the summability of $(J(n))$, and $|J(n)| \leq C n^{-2-\varepsilon}$, we obtain

$$|f(x) - f(y)| \leq Cr + C \sum_{n \geq 1} n^{-2-\varepsilon} \min\{1, 2^{n+1} r\}. \quad (7)$$

For $0 < r < 1/4$, split the sum at $M = \lfloor \log_2(1/r) \rfloor$. The part $n \leq M$ is at most

$$Cr \sum_{n \leq M} 2^n n^{-2-\varepsilon} \leq CM^{-2-\varepsilon},$$

whereas the tail is at most $CM^{-1-\varepsilon}$. Thus

$$|f(x) - f(y)| \leq C\omega_{1+\varepsilon}(r). \quad (8)$$

In particular $f \in C^{\varepsilon \log}(X)$. This argument works unchanged for every coefficient sequence in (2).

The natural synchronous coupling prepends the same random point a to both sequences. It has the exact contraction

$$d(ax, ay) = \frac{1}{2}d(x, y). \quad (9)$$

Along a coupled inverse path, (8) and the elementary integral comparison give, uniformly in its length,

$$\sum_{j \geq 0} \omega_{\varepsilon}(2^{-j}r) \leq C \int_0^{\infty} (\log(r_0/r) + t \log 2)^{-\varepsilon} dt \leq C' \omega_{\varepsilon-1}(r). \quad (10)$$

Hence f is flat with respect to $\omega_{\varepsilon-1}$ and the synchronous coupling.

Kloeckner's corrected normalization theorem applies to backward transition kernels on arbitrary compact metric spaces; it is not restricted to finite fibers. It gives the positive h , ρ , and a cohomologous normalized potential $\bar{f}$ which remains flat. The synchronous coupling is Markovian, non-dilating, and contracts with probability one by (9). The corrected stable-decay criterion therefore gives weighted-coupling decay of polynomial degree $\varepsilon - 1$. The core Wasserstein contraction theorem, followed by the standard observable/measure duality, is precisely (5). Notice that this duality supplies $\|\varphi\|_{(\varepsilon-1)\log}$ on the right side.

Finally $\varepsilon > 2$, so $\sum_n n^{-(\varepsilon-1)} < \infty$. Therefore

$$v = \sum_{n=0}^{\infty} \mathcal{L}_{\bar{f}}^n \varphi$$

converges uniformly and hence in $L^2(\mu_f)$; telescoping gives the Poisson equation $(I - \mathcal{L}_{\bar{f}})v = \varphi$. The FCLT theorem invoked in the source now applies. $\square$

Scope and verification note

This packet records a full literature answer for the precise compact-sphere, long-range question. It does not claim a new proof of Kloeckner's general coupling theorem. The consolidated arXiv version of that theorem must be used: it incorporates the 2023 corrigendum by strengthening the stability hypothesis. Condition (9) satisfies the corrected criterion, so the corrigendum creates no gap here.

The 2023 answer preprint contains two harmless presentational slips. First, its proof of logarithmic regularity writes a coordinate estimate without the factor depending on the coordinate; the split estimate (6)–(8) is the correct replacement. Second, one displayed uncountable-alphabet estimate has a potential norm where the observable norm is required; (5) is the form yielded by Wasserstein duality. Neither affects the claimed $O(N)$ FCLT.

References

- [1] L. Cioletti, L. Melo, R. Ruviano, and E. A. Silva, *On the dimension of the space of harmonic functions on transitive shift spaces*, arXiv:2009.07209; Adv. Math. 385 (2021), 107758.

- [2] E. A. Silva, E. G. Mesquita, and E. Matias, *Some Probabilistic Properties of General Topological Markov Chains*, arXiv:2310.01697, 2023.
- [3] B. R. Kloeckner, *An optimal transportation approach to the decay of correlations for non-uniformly expanding maps*, Ergodic Theory Dynam. Systems 40 (2020), 714–750; corrigendum, *ibid.* 43 (2023), 3538–3544.